GERMÁN KÜBER

Senior Blockchain Engineer — Rust, Node.js / TypeScript, MEV & Multi-Chain Infrastructure

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PROFESSIONAL SUMMARY

Senior backend and blockchain engineer with 15+ years in distributed systems and 4+ years shipping production blockchain infrastructure. Polyglot across **Rust (tokio, async), TypeScript / Node.js, Python / Cython and .NET**, building and operating high-volume on-chain transaction systems across EVM (Ethereum, Arbitrum, Optimism, Base) and non-EVM chains (Solana, StarkNet, Polkadot). At Borderless, operated MEV searcher pipelines processing $50M+ in daily on-chain volume, integrating directly with DeFi protocols — AMMs (Uniswap V2/V3, Curve, Raydium, PumpSwap) and lending markets (Aave) — using **ethers.js, viem, web3.js and @solana/web3.js** alongside Rust-side execution. Deep hands-on knowledge of transaction lifecycles, consensus mechanisms, smart-contract interaction, on-chain data and cryptographic primitives. Microservices in production CI/CD (Kubernetes, AWS/Azure). Active open-source contributor through 80+ international talks and Net-Baires (100K+ members).

CORE EXPERTISE

MEV Infrastructure: MEV searcher pipelines, transaction delivery, validator / TPU routing, mempool analysis, sandwich attacks, arbitrage detection, liquidation detection, gas optimization, sub-slot execution. Flashbots stack: MEV-Boost, Suave, MEV-Share, PBS (Proposer-Builder Separation), relays. Jito stack: bundles, BAM (Block Assembly Marketplace), BuilderNet, Jito-Solana, Agave

Blockchain & Web3: EVM (Ethereum, Arbitrum, Optimism, Base), Solana, StarkNet, Polkadot SDK / Substrate / FRAME / XCM, parachains, runtime upgrades, smart contracts, account abstraction (ERC-4337), cross-chain flow modeling, DEX integration (Uniswap V2 / V3, Curve, Raydium, PumpSwap), lending markets (Aave), Ethereum clients (Geth reverse-engineering, familiar with Reth, Lighthouse, Prysm)

Programming Languages & Smart Contracts: Rust (tokio, async), TypeScript / Node.js, Solidity, Cairo, Python, Cython, Go, JavaScript, C# / .NET

Web3 Tooling & Libraries: ethers.js, viem, web3.js, @solana/web3.js, Foundry / Forge, Anchor (Solana), Hardhat

Distributed Systems & Networking: Low-latency networking (sub-10ms execution), async runtimes (tokio), concurrency, multi-threaded systems, fault tolerance, high-throughput pipelines, P2P networking, consensus protocols (hands-on with GRANDPA / BABE, familiar with Tendermint and Raft), gRPC, Protocol Buffers, event-driven architectures

Cryptography & Zero-Knowledge: ZK proofs (Noir, Barretenberg, Garaga on StarkNet, Cairo), ZK-SNARKs / STARKs, account abstraction (ERC-4337), digital signatures, hashing, Merkle trees, cryptographic primitives applied to MEV, on-chain monitoring, and verifiable computation

Observability & Tooling: Prometheus, OpenTelemetry, tracing crate, custom Rust telemetry (50+ real-time production metrics), structured logging, post-mortems, on-call discipline

AI / LLM Systems: LLM orchestration (OpenAI, Anthropic, Gemini), LangChain, LangGraph, MCP (Model Context Protocol), RAG pipelines, multi-agent architecture, real-time audio pipelines, multi-provider routing

Cloud & Infrastructure: AWS, Azure, Docker, Kubernetes, FastAPI, SignalR, CI / CD pipelines, observability stacks, Linux internals

Engineering Leadership: Technical leadership for blockchain and distributed systems teams, hiring and technical interviewing for Rust / blockchain hires, code review and mentoring, technical roadmap, architectural decision records (ADRs), on-call and incident response, open-source contribution practices

PROFESSIONAL EXPERIENCE

Lead Software Engineer (Blockchain / MEV)

Borderless Capital / CTF Capital — Aug 2021 – Apr 2025 — Remote

- Shipped and operated production MEV searcher infrastructure in Rust (tokio, async) and Python / Cython across EVM (Ethereum, Arbitrum, Optimism) and Solana, owning architecture, code review, mentoring, and performance optimization while remaining hands-on in production code.
- Built Node.js / TypeScript services and tooling that interact with DeFi protocols on-chain — AMMs (Uniswap V2/V3, Curve, Raydium, PumpSwap) and lending markets (Aave) — using ethers.js, viem, web3.js and @solana/web3.js for protocol analysis, on-chain data reads and transaction submission.
- Integrated with the Flashbots stack — MEV-Boost, Suave, MEV-Share, PBS, and relays — for Ethereum MEV execution and bundle submission.
- Integrated with the Jito ecosystem on Solana — bundles, BAM (Block Assembly Marketplace), BuilderNet, Agave, and the Jito-Solana fork — for sub-slot transaction inclusion and validator routing.
- Pipelines processed 10K+ opportunities per day from $50M+ in daily on-chain flow and contributed $100K+ in attributable annual P&L with ultra-low-latency execution under 10ms p99.
- Built custom transaction delivery infrastructure with parallel broadcasting across RPC endpoints and direct validator / TPU routes, achieving sub-slot Solana inclusion (~400ms).
- Designed strategy logic for liquidation detection, multi-hop arbitrage across 5+ DEXs, and cross-chain flow modeling across EVM and Solana.
- Reverse-engineered Geth (Go-Ethereum) internals to inform Rust-side designs for transaction simulation and gas estimation in internal trading tools.
- Engineered low-latency observability with Prometheus and custom Rust telemetry, instrumenting 50+ real-time metrics across slot timing, validator rotation, and execution outcomes.
- Owned 24/7 production reliability and incident response for trading-critical pipelines, including on-call rotation, post-mortems, and observability ownership.

Lead AI Architect & Co-Founder
KliverAI (Own Venture, closed Apr 2026) — Apr 2025 – Apr 2026 (full-time) — Remote, Spain

- Designed a Zero-Knowledge proof system using Noir, Barretenberg, and Garaga on StarkNet for verifiable expert credentials in a decentralized evaluation marketplace.
- Owned technical strategy, architectural decisions, and delivery as the senior-most engineer; led platform development end-to-end on a polyglot stack (Node.js / TypeScript, Python, .NET, Kubernetes, AWS).
- Architected real-time audio pipelines using OpenAI Realtime API and SignalR, achieving sub-200ms streaming latency across concurrent user sessions.
- Built an adaptive simulation engine with multi-agent architecture on LangGraph, including multi-step branching logic and RAG-powered personalization.
- Integrated three LLM providers (OpenAI, Anthropic, Gemini) with dynamic model routing for cost and performance optimization.
- Hired and led a distributed team of up to 3 engineers, conducting technical interviews and final hiring decisions.

Fractional CTO (Part-time, hands-on)
Rocking Product — Apr 2025 – Nov 2025 (part-time, parallel to KliverAI) — Remote / Buenos Aires

- Led technical roadmap and platform direction for an early-stage product, inheriting an 8-person engineering team and partnering directly with founders to translate business priorities into delivery.
- Shipped multiple production products on a Node.js / TypeScript and Python (FastAPI, LangChain) stack, establishing the team's first LLM evaluation and monitoring practices.
- Optimized LLM usage and cloud infrastructure through model selection, prompt compression, and semantic caching, cutting combined API and cloud spend by 45%.
- Proposed and led a full-stack migration from a fragmented JavaScript codebase to .NET, modernizing the architecture for long-term scalability.

Senior Software Architect (Full-Stack)
Paramo Technology — Aug 2020 – Aug 2021 — Remote / Argentina

- Held company-wide architecture authority for a large-scale online casino platform — part of and accountable for every cross-team technical decision across 9+ engineering teams (200+ professionals, each with its own coordinator), working directly with the CTO and defining organization-wide engineering standards.
- Designed and rolled out event-driven microservices using .NET handling real-time game state, wallet operations, and regulatory reporting with strict consistency requirements.
- Created and delivered internal training programs on architecture patterns, testing discipline, and operational best practices.
- Partnered with product, operations, and infrastructure teams to align technical roadmap with business growth, regulatory constraints, and platform reliability.

Software Architect (Full-Stack)
CloudX (Software Consultancy) — Aug 2017 – Aug 2020 — Buenos Aires

- Led architecture across multiple enterprise product lines as part of a software consultancy, designing Angular/React frontends and Node.js / TypeScript and .NET microservices backends using Domain-Driven Design, delivered for a range of clients.
- Delivered technical training and conference talks to partners and internal teams, and collaborated directly with clients on-site for requirements gathering and technical adoption.

Software Architect (Angular / .NET)
Majestic Hotel Group — Jan 2014 – Aug 2017 — Argentina

- Designed and shipped full-stack features in Angular and .NET for internal hotel management platforms; led the migration from AngularJS to Angular 4.
- Redesigned the data model and refactored backend workflows toward an event-oriented architecture, reducing coupling across integrations.

.NET Web Developer
Plus54 — Aug 2009 – Mar 2014 — Buenos Aires

- Early career: built .NET web applications, internal tooling, and mobile components end-to-end; integrated external APIs with unit testing and reliable data exchange patterns.

KEY ACHIEVEMENTS

MEV Infrastructure at Borderless — Rust and Cython pipelines processing $50M+ in daily on-chain volume and contributing $100K+ in attributable annual P&L.
Polkadot Builder Program Graduate — Completed Polkadot Blockchain Academy in Lisbon, Portugal (2026).
Microsoft MVP, Developer Technologies — 8 consecutive years (2017–present), recognized for technical content and community contributions.
80+ International Technical Talks — .NET Conf, DevFest, NetCoreConf, vOpen, Global Azure Bootcamp, and meetups across LATAM and Europe.
Net-Baires Founder — Founded and grew Argentina's largest .NET developer community to 100K+ members across 10+ years.
Microsoft Student Partner — 5 years (2016–2021).

EDUCATION & CERTIFICATIONS

Microsoft Certified: Azure Solutions Architect Expert (AZ-305) — Microsoft
Microsoft Certified: Azure Associate (AZ-104 / AZ-204 level) — Microsoft
Polkadot Builder Program — Polkadot Blockchain Academy, Lisbon, Portugal (2026)
Systems Engineer — Universidad Abierta Interamericana (UAI), Buenos Aires, Argentina (2017–2021)
Microsoft MVP, Developer Technologies — Feb 2017 – Present (8 consecutive years)

OPEN SOURCE & COMMUNITY LEADERSHIP

Net-Baires (.NET Community) — Founder (2015–present). Argentina's largest .NET developer community, 100K+ members, 10+ years of monthly meetups and conferences.

Crypto Para Mortales — Founder (2023–present). Spanish-language blockchain and crypto education conference.
Patrones Para Mortales — Founder (2020–present). Educational content brand on software design patterns and architecture.
YouTube channel (youtube.com/@germankuber) — technical deep-dives on Rust, blockchain, AI, and software architecture.

SPOKEN LANGUAGES

Spanish: Native — English: Full Professional Proficiency — daily use in international remote teams